



Explainable Active Learning (XAL): Toward AI Explanations as Interfaces for Machine Teachers

Ghai et al. (2020)

Presented by *Lucia Gordon, Matthew Nazari, and Catherine Yeh*
March 22, 2023

Prediction Task

How much money might a person with the following attributes make in a year?

- Dataset: Adult Income
- Binary prediction: whether the annual income of an individual is $>$ or $<$ \$80k
- Linear model: logistic regression with L2 regularization

Attributes	Values
Age	39
Workclass	Private
Years of Education	13
Marital status	Never married
Occupation	Executive & managerial
Race	White
Gender	Male
Hours per week	50

Customer profile

Active Learning

- Current ML development processes are bottlenecked by
 - resources and expertise
 - how, where, and when we can leverage human knowledge



Active Learning in Prediction Task

- Participants judge the income level of data points queried by the model
- Query method: uncertainty sampling
- The annotations could be faulty
- BUT... explanations might be able to help reveal faulty model beliefs

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Problem Statement

- Active learning interfaces remain minimal and opaque
 - annotator cannot monitor training progress
 - annotator is unaware of the effectiveness of their teaching
- Explanations show potential for better active learning

The top right corner of the slide features two overlapping squares: a light blue square on top and a dark blue square below it, both of which are partially cut off by the right edge of the frame.

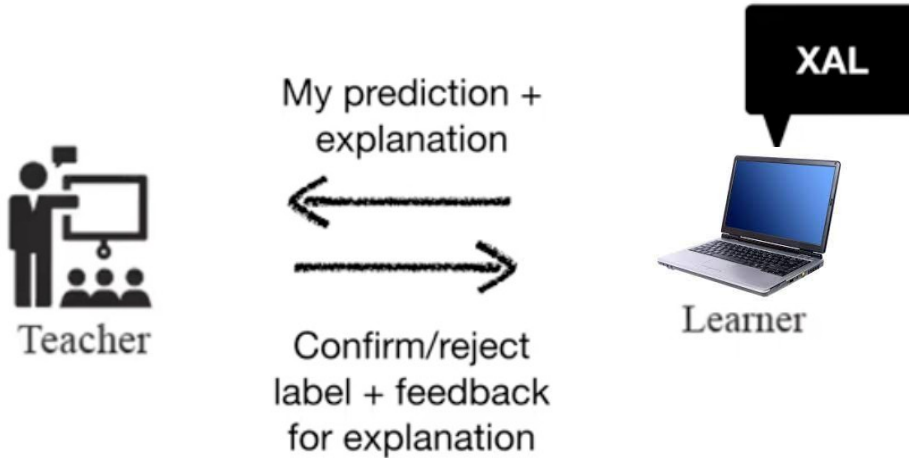
Big Question:

Can explanations improve the active learning process?

Summary of Contributions

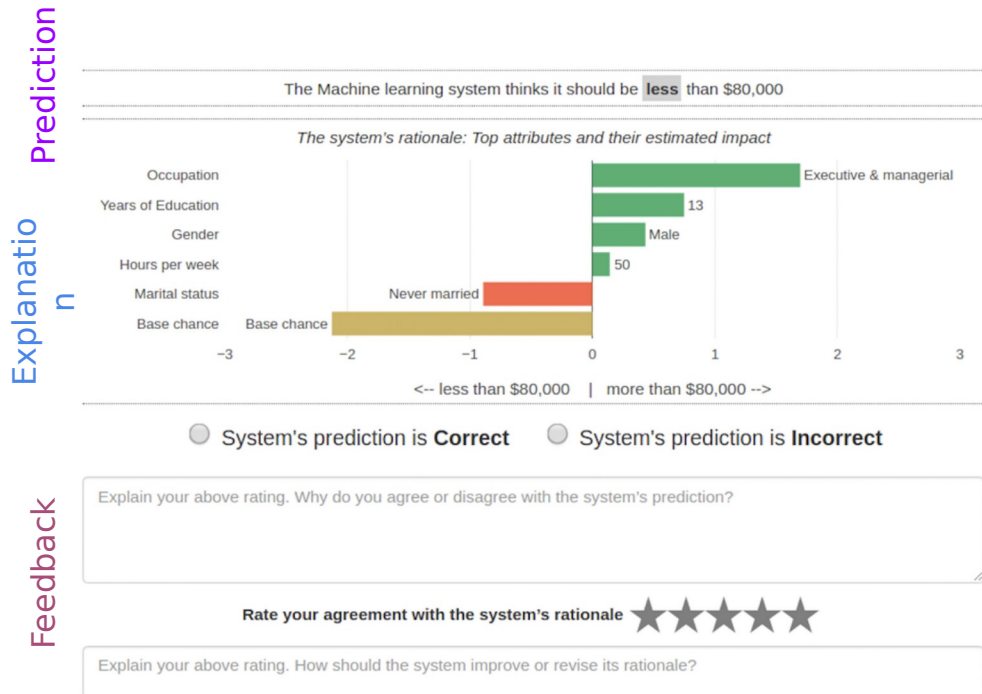
1. Provide insights into explainable AI as an interface for active learning
2. Propose a novel active learning paradigm: *explainable active learning* (XAL)
3. Conduct an empirical study using the prediction task we described to investigate the impact of explanations on annotation experience

Explainable Active Learning (XAL)



- Mimics how humans teach and learn
- Explanations can make it easier to reject an incorrect prediction
- Explanations can inform the feedback

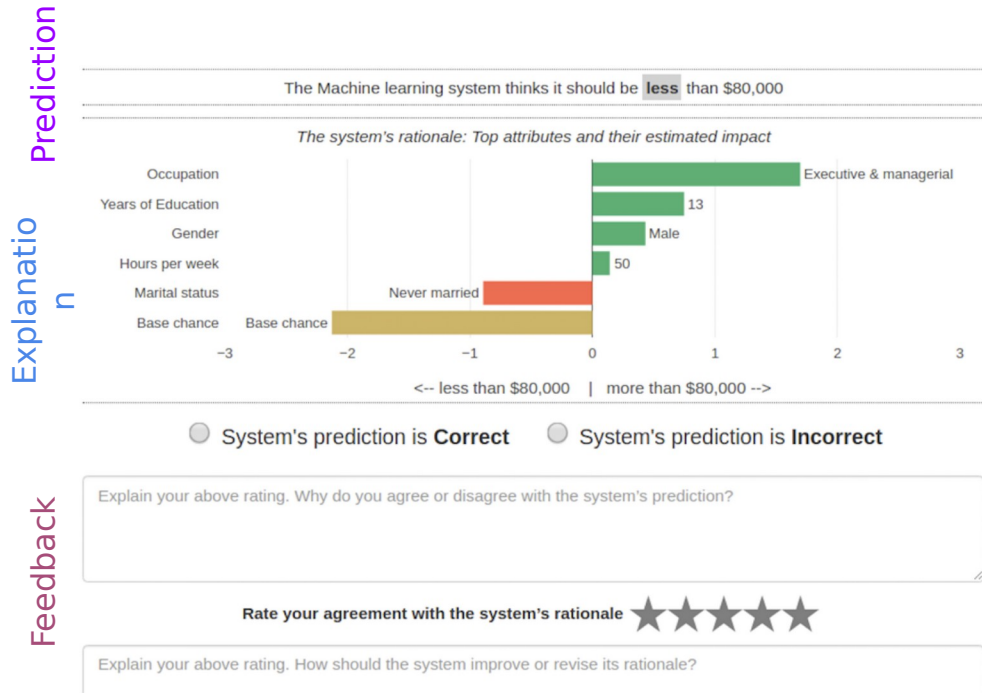
Presentation of Prediction + Explanation



Design choices

- Explain AL with local feature importance
- Present local feature importance with visualization
- Features were sorted by their importance
- Top 5 most important features were included in the visualization
- Green = positive features
- Red = negative features
- Base chance = model intercept

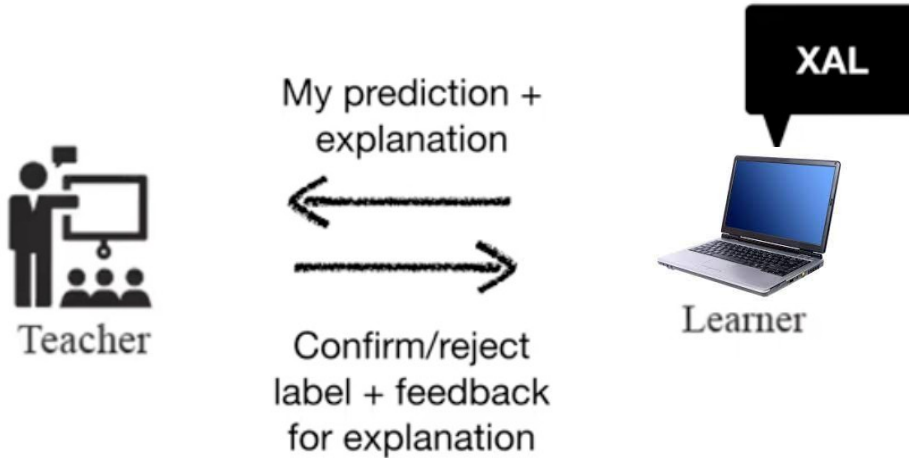
Presentation of Prediction + Explanation



How it differs from previous work

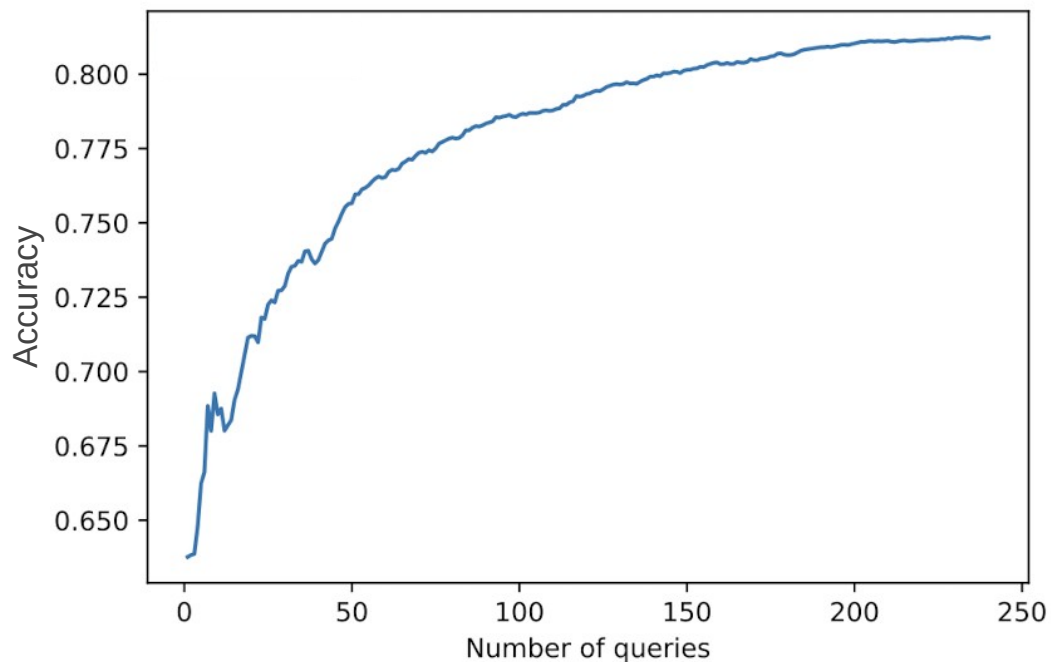
- Presents the model's reasoning rather than requesting global feature weights
- Goes beyond text-based models

Explainable Active Learning (XAL)



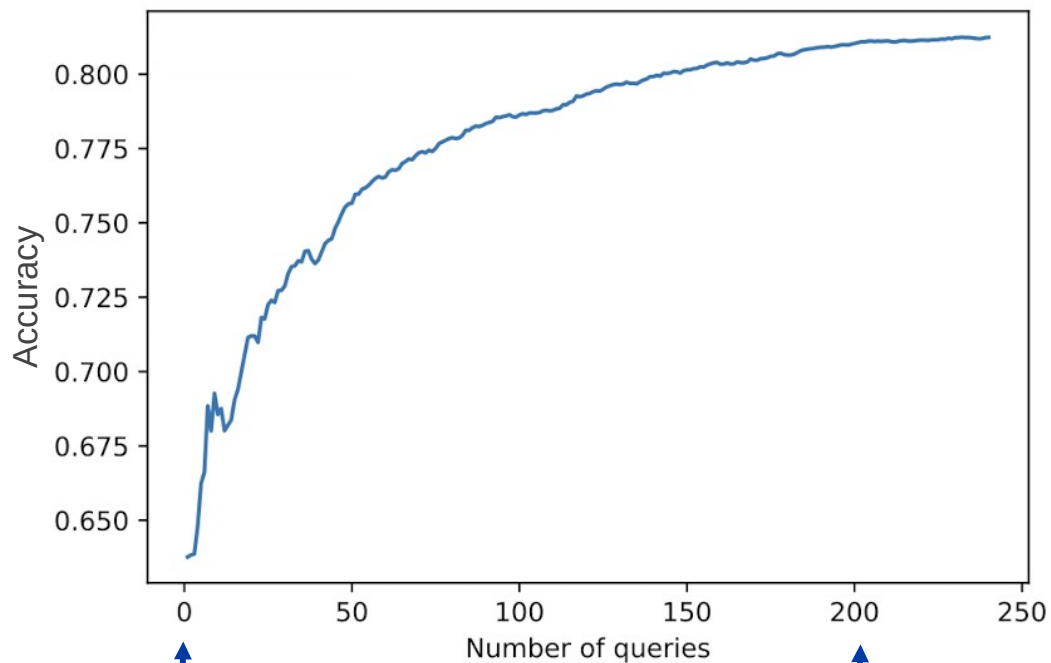
They design a study to explore **how people naturally want to teach a model with explanations** in the form of a local feature importance visualization

Active Learning Simulation



- Wanted to examine the effect of explanations at different stages of AL
- Used a simulation to define “early-stage” and “late-stage” models
- Queried instances were annotated using ground-truth labels in the simulation

Active Learning Simulation



Early
stage

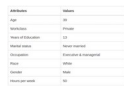
Late
stage

- Early stage → 0 queries
- Late stage → 200 queries (where accuracy plateaus)

Experimental Design

- Amazon Mechanical Turk: 36 participants
- Three **conditions** (each participant assigned to one)

a. AL: customer profile presented



Field	Value
Name	John Doe
Age	35
Gender	Male
Occupation	Software Engineer
Income	\$75,000
Marital Status	Married
Number of Children	2
Home Ownership	Owns
Years in Current Home	5
Years in Current City	10
Years in Current Country	15
Years in Current State	10
Years in Current Zip	5

b. CL: customer profile + model prediction presented



Field	Value
Name	John Doe
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+

The Machine learning system thinks it should be **less** than \$80,000

c. XAL: customer profile + model prediction



Field	Value
Name	John Doe
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+

The Machine learning system thinks it should be **less** than \$80,000

+

+ explanation



- Two learning **stages** (each participant completed both)

a. Early

b. Late

- 20 annotations per experiment (condition + stage)
- Domain knowledge training (statistics, practice trials, \$2 bonus for consistency with ground-truth)

Research Question 1

How do local explanations impact the annotation and training outcomes of AL?

Results: Annotation + Learning Outcomes

Table 1. Results of model performance and labels

Stage	Condition	Acc.	Acc. improve	F1	F1 im- prove	%Agree	Human Acc.
Early	AL	67.0%	13.7%	0.490	0.104	55.0%	66.7%
	CL	64.2%	11.7%	0.484	0.105	58.3%	62.1%
	XAL	64.0%	11.8%	0.475	0.093	62.9%	63.3%
Late	AL	80.4%	0.1%	0.589	0.005	47.9%	54.2%
	CL	80.8%	0.2%	0.587	0.007	55.8%	58.8%
	XAL	80.3%	-0.2%	0.585	-0.001	60.0%	55.0%

Compare to initial model performance (before 20 queries)

RQ1: Overall, human accuracy and % agreement decrease in the later stage, while model accuracy increases.

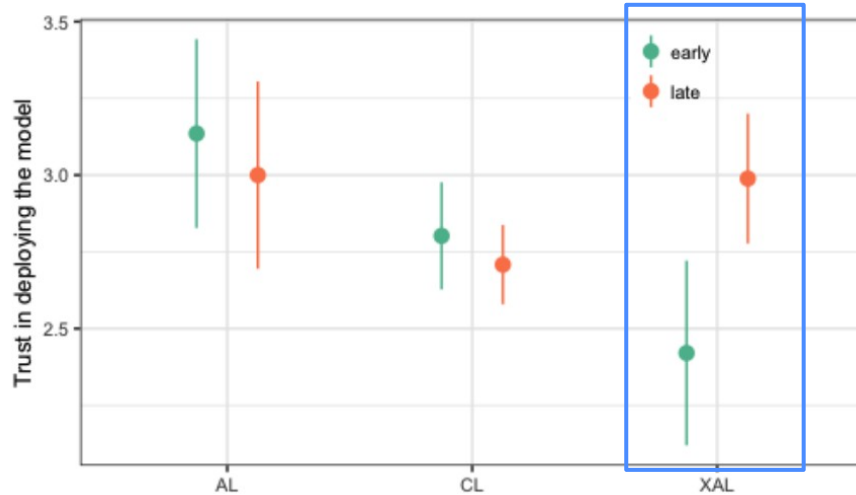
Research Question 2

How do local explanations impact annotator experiences?

Hypotheses

1. Explanations support ***trust calibration***
2. Explanations improve ***annotator satisfaction***
3. Explanations increase perceived ***cognitive workload***

Results: Annotator Experience



H1: Explanations support trust calibration

RQ2: Trust in XAL condition is low early on, but increases with later stage model.

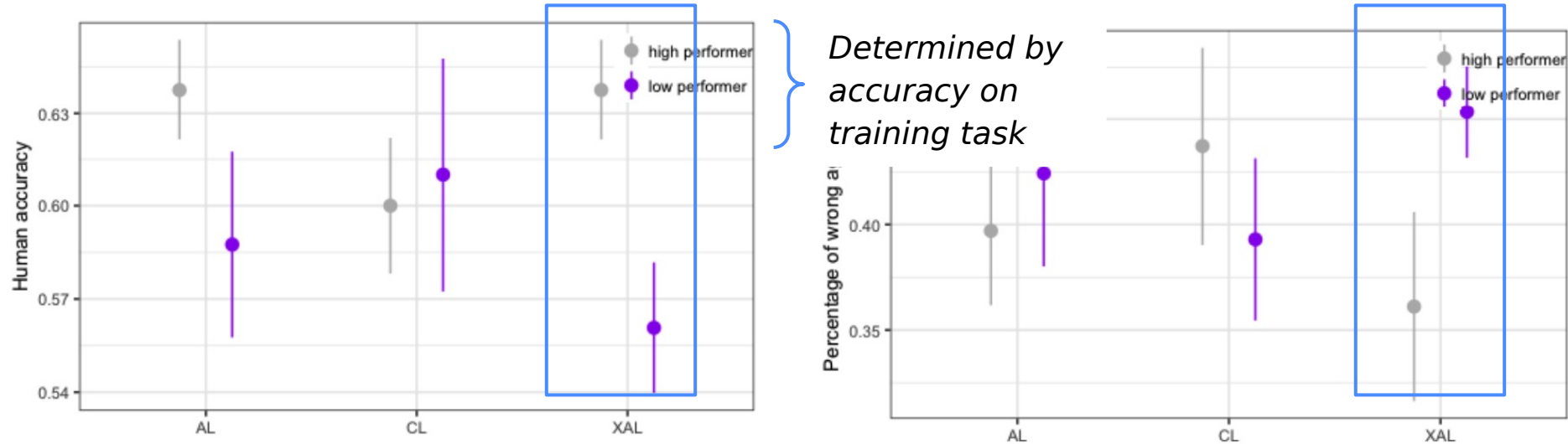
Research Question 3

How do individual factors, specifically ***task knowledge***, ***AI experience***, and ***need for cognition***, impact annotation and annotator experiences with XAL?

Hypotheses

4. Annotators with lower task knowledge benefit more from XAL
5. Annotators inexperienced with AI benefit more from XAL
6. Annotators with lower need for cognition have a less positive experience with XAL

Results: Annotation + Learning Outcomes



RQ3: Participants with less task knowledge had lower accuracies and higher blind trust in XAL condition.

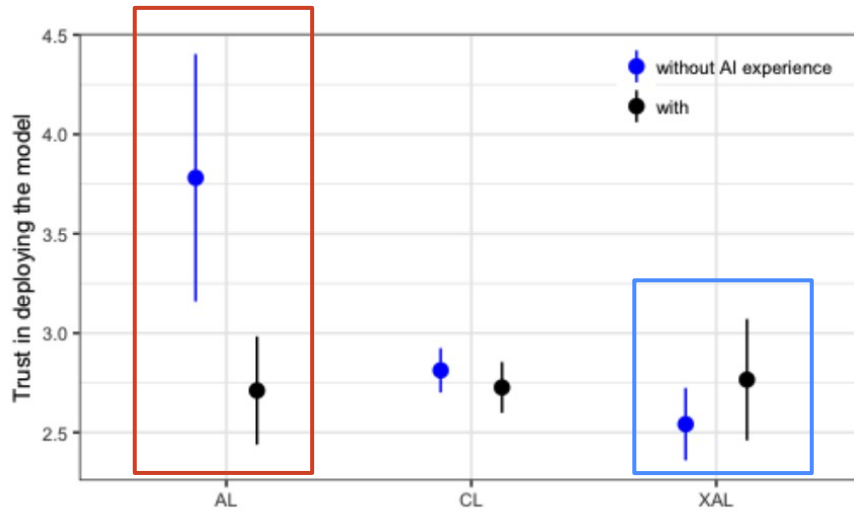
Results: Annotation + Learning Outcomes



H4: Annotators with lower task knowledge benefit more from XAL

RQ3: Participants with less task knowledge had lower accuracies and higher blind trust in XAL condition.

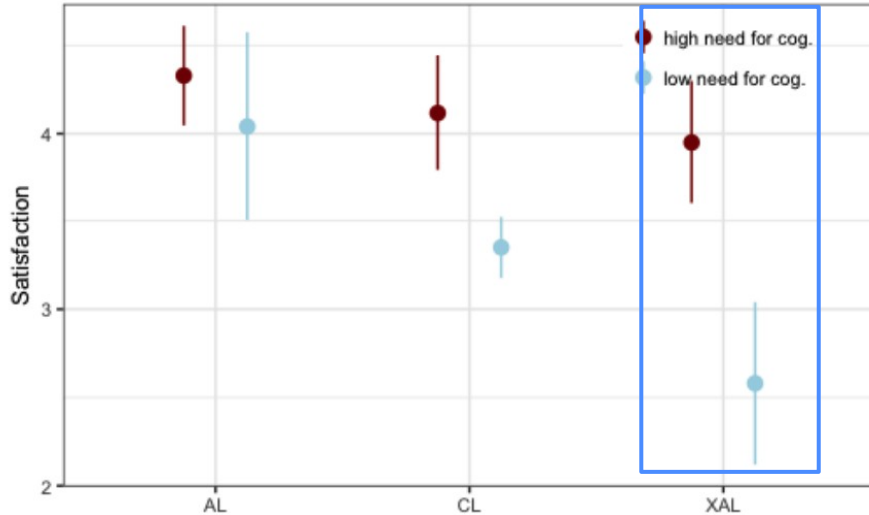
Results: Annotator Experience



H5: Annotators inexperienced with AI benefit more from XAL

RQ3: Explanations helped calibrate trust for those without AI experience.

Results: Annotator Experience

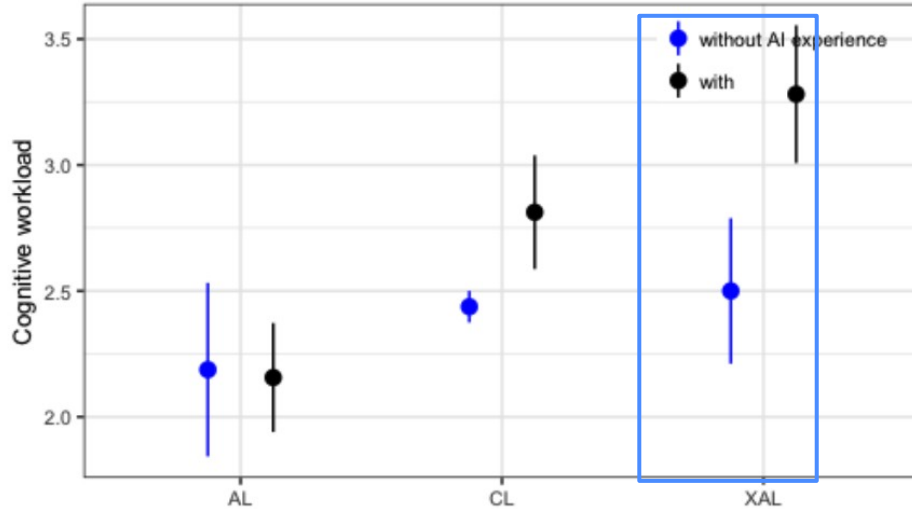


H2: Explanations
improve annotator
satisfaction

H6: Annotators with lower
need for cognition have a
less positive experience
with XAL

RQ3: Explanations have a significant negative effect on satisfaction for those with low need for cognition.

Results: Annotator Experience



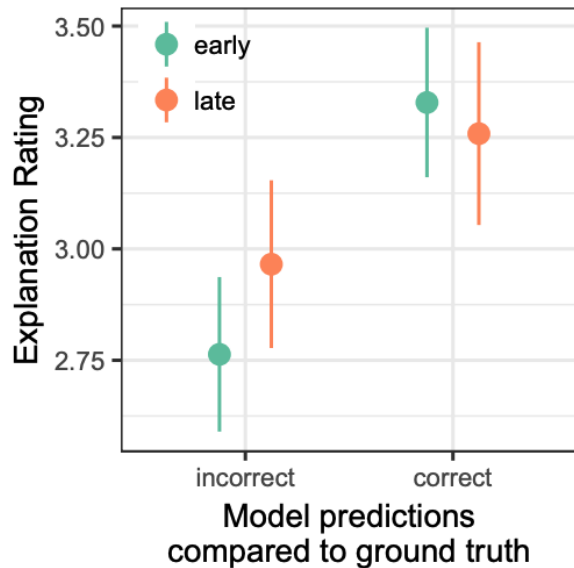
H3: Explanations increase perceived cognitive workload

RQ3: XAL induces significantly higher workload for those with AI experience.

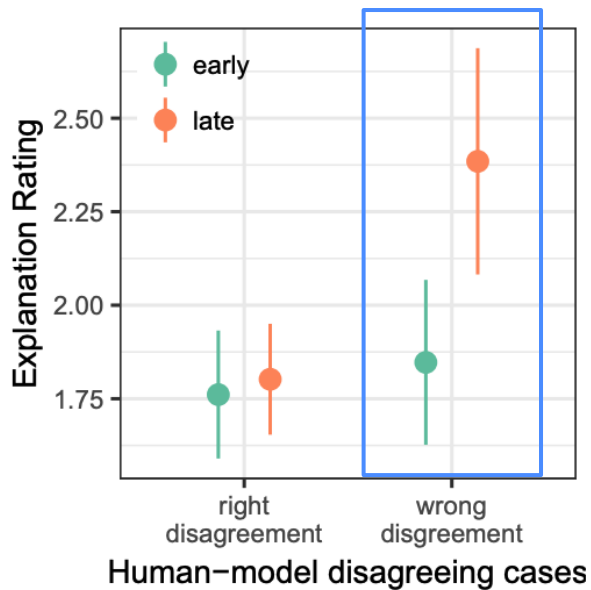
Research Question 4

What kind of feedback do annotators naturally want to provide upon seeing local explanations?

Results: Feedback for Explanation



Explainable (XAL) condition

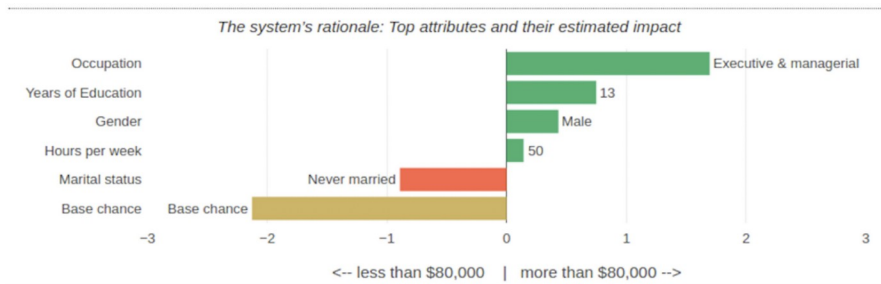


RQ4: Explanation ratings are higher when model is correct and when annotators wrongly disagree with the model in later stage tasks.

Results: Feedback for Explanation

Open Form Feedback

- Tuning weights (N = 81)
- Removing, changing direction of, or adding features (N = 28)
- Ranking or comparing multiple feature weights (N = 12)
- Reasoning about combination and relations of features (N = 10)
- Logic to make decisions based on feature importance (N = 6)
- Changes of explanation (N = 5)



Related Work: Active Learning

- Many AL paradigms have been developed, and the paper lists many
- Very little or no attention has been given to understanding or improving how humans interact with AL algorithms
 - In human-robot interaction, a natural language interface was developed (Cakmak et. al)
- They identify the need to study annotation interactions with real-time AL algorithms



Related Work: Interactive ML

- AL is sometimes considered to be within interactive machine learning (iML)
- iML approaches value transparency over performance
- There have been empirical studies demonstrating iML techniques lower need for data, but little else
- iML approaches are esoteric, but explanations as interfaces could help non-ML experts



Related Work: Explainable AI

- The paper describes explainable AI (XAI) from a broad overview
- Recent XAI studies find users' understanding of AI systems improved by explanations
- There are still many unknowns to how explanations could reduce knowledge barriers to train ML models

Future Work

Explanations for Active Learning

- Alternative designs to mitigate anchoring effect

CSCW '21

IS

To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making

ZANA BUÇINCA, Harvard University, USA

MAJA BARBARA MALAYA, Lodz University of Technology, Poland

KRZYSZTOF Z. GAJOS, Harvard University, USA

The AI is 87% confident in its suggestion

See AI's suggestion ▼



The AI is processing the image

(b) uncertainty (SXAI)

(c) on demand (CFF)

(d) wait (CFF)

Future Work

Learning from explanation based feedback

- Scaffolding the elicitation of high-quality, targeted feedback

CSCW '15

Structuring, Aggregating, and Evaluating Crowdsourced Design Critique

Kurt Luther¹, Jari-Lee Tolentino², Wei Wu³, Amy Pavel³,
Brian P. Bailey⁴, Maneesh Agrawala³, Björn Hartmann³, Steven P. Dow¹

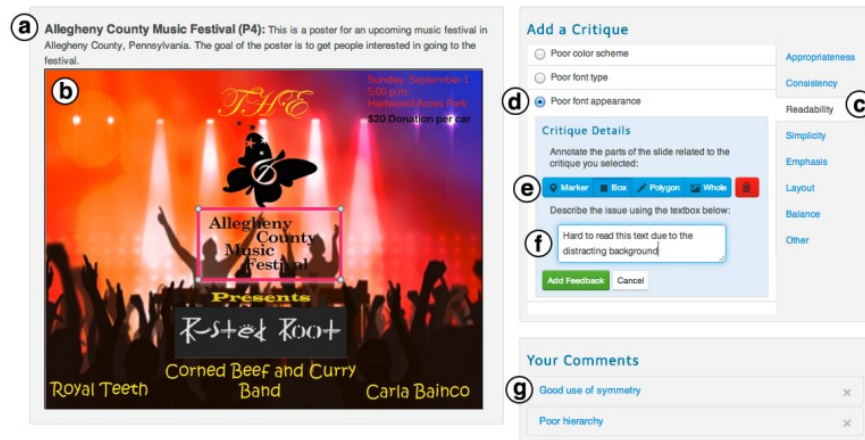
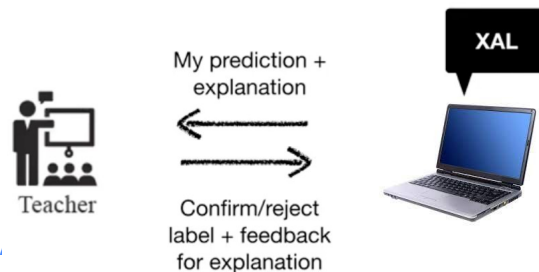


Figure 3. The CrowdCrit critique interface.

Conclusions

Key contributions

- New approach for machine teaching: $XAI + AL = X$.



Limitations

- If domain experts are usually the ones doing the labeling, why not survey them?
- Small number of participants
- Only tried uncertainty sampling (but other strategies may have different impacts on annotator experiences)
- Only tried one explanation method with a highly interpretable model - how would the results change under different explanation methods or more black-box models?
- What if the number of relevant features is larger (such as pixels in an image?)

Discussion Questions

- Has anyone used AL in the past? If so, do you think adding explanations would have been helpful? Why or why not?
- What applications of XAL are you most excited about? Or, any particular scenarios that you think it would work especially well/poorly in?
- Do you think XAL is feasible with less inherently interpretable models or other explanation methods? How do you think the annotations and the annotation experience would change?

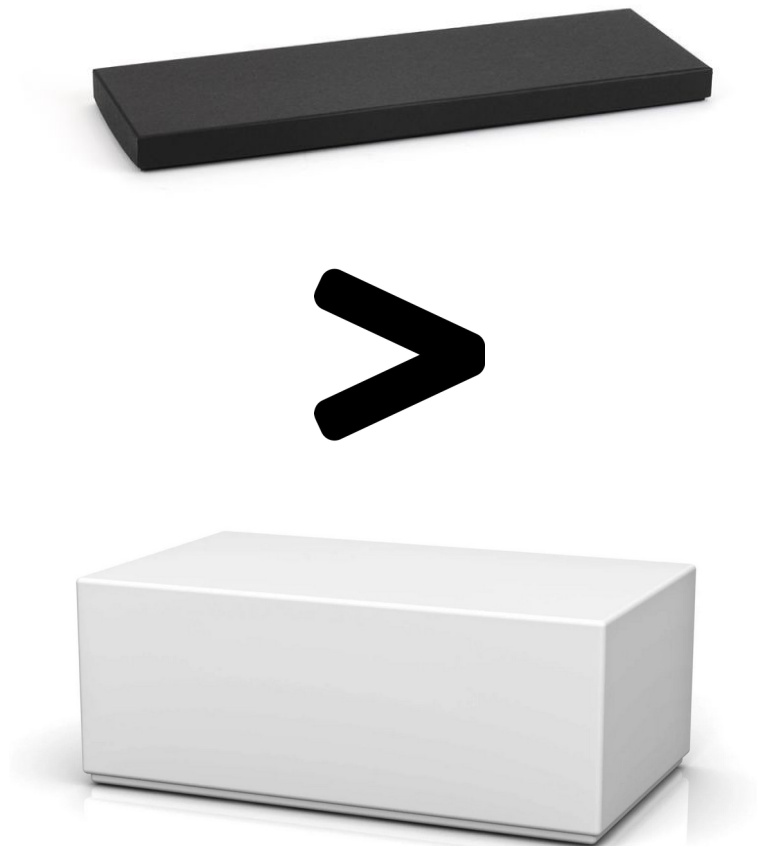
Explaining Machine Learning Models with Interactive Natural Language Conversations Using **TalkToModel**

Authored by Dylan Slack, Satyapriya Krishna, Himabindu Lakkaraju, and Sameer Singh

Presented by Oam Patel, Jason Wang, and Lucas Monteiro Paes

Motivation

- Simple and intuitive explanations for ML models is a bottleneck to adoption
- Flexibility and accuracy tradeoff for inherently-interpretable models
- Using post-hoc methods is difficult empirically (which explanations, how to interpret, follow-up questions, etc)



Related Work

1. Language-interpretability Tool (LiT)

- Open-source platform for understanding NLP models
- Uses local explanations (ala LIME), aggregate data statistics, and counterfactual explanations

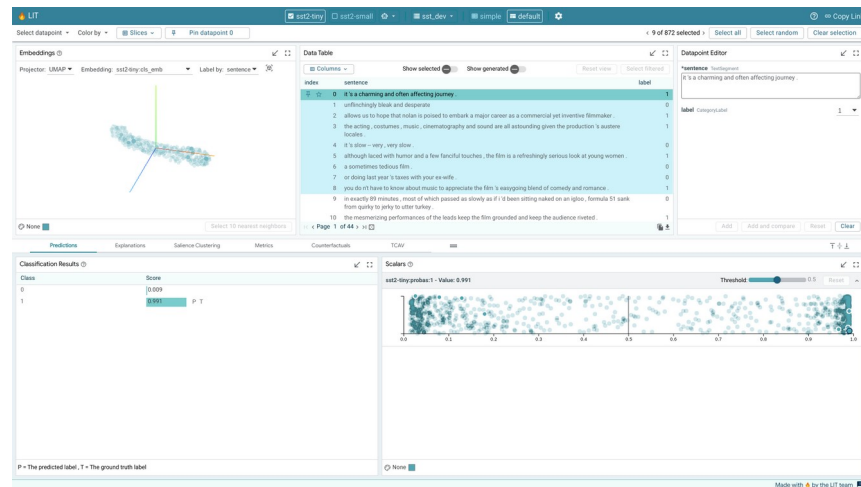
1. “What-If” Tool

- Helps users perform counterfactual analysis for models

1. *explainerdashboard*

- Used as a baseline for experiments

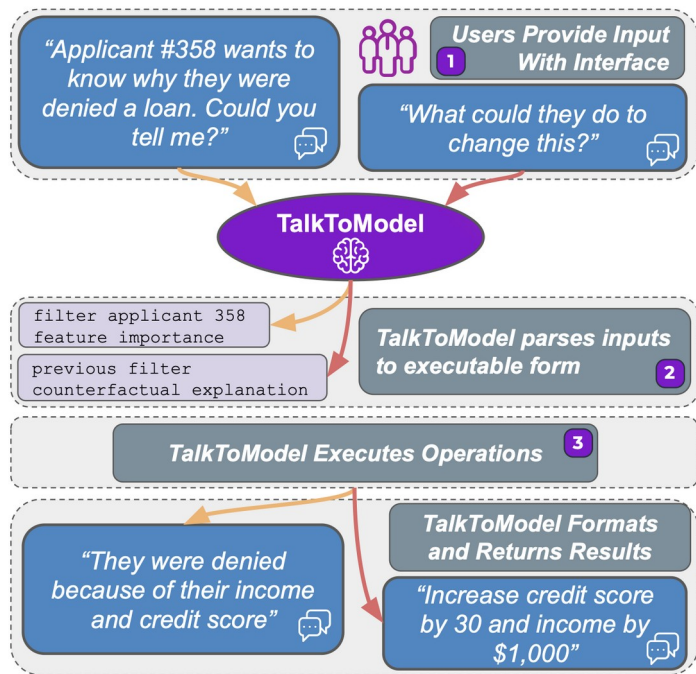
Unfortunately, relatively high barrier to entry and no follow up questions



Problem Statement

- Design a system that makes it easy for lay practitioners to apply post-hoc interpretability methods to black-box models
- Desiderata
 - Dialogue system that can handle many conversation topics (general data trends, questions about specific predictions, etc)
 - Usable for a variety of data types and model classes (i.e. treatment prediction, risk of relapse, interest rate calculations, etc)
 - Doesn't require a high level of expertise

Summary of Contributions

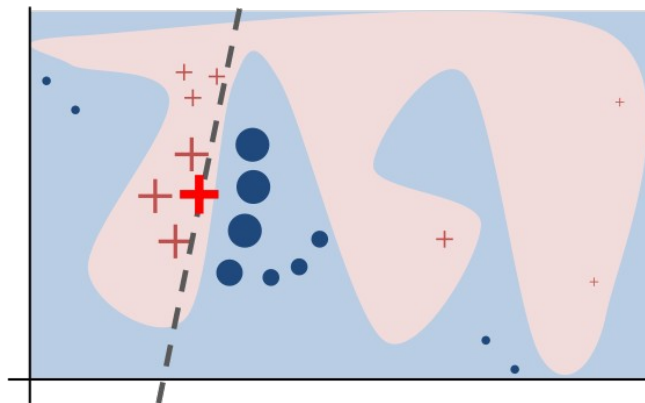


- Introduce TalkToModel which enables open-ended dialogue for understanding a given dataset+classifier pair
 - Why a prediction occurred, how it would change if data changed, how to flip predictions, general statistics regarding the data distribution, etc
- Three parts
 - Dialogue engine (LLM backend)
 - Execution engine (run many explanations, pick the best one)
 - Text interface (to enable conversations)

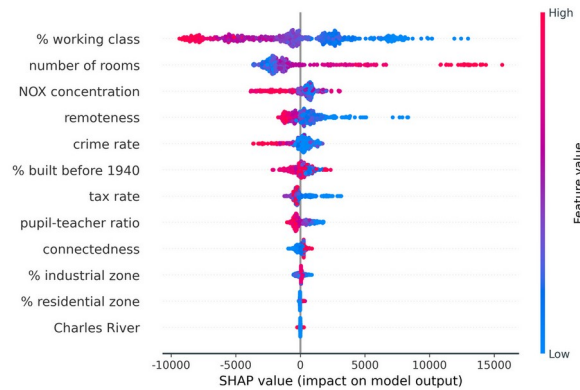
Demo Time!

Method

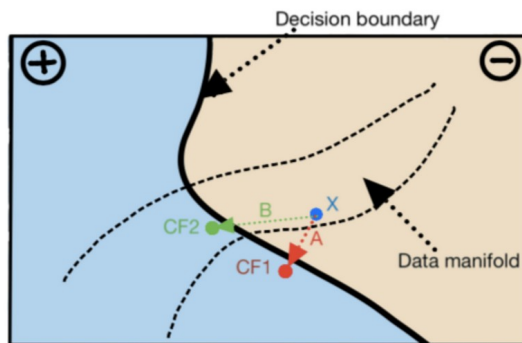
Background



Lime Illustration

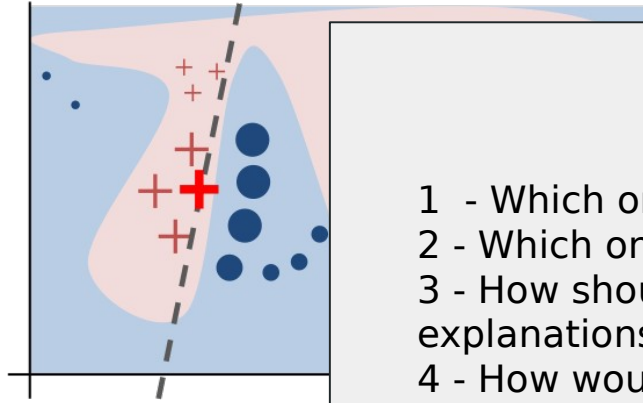


SHAP Illustration



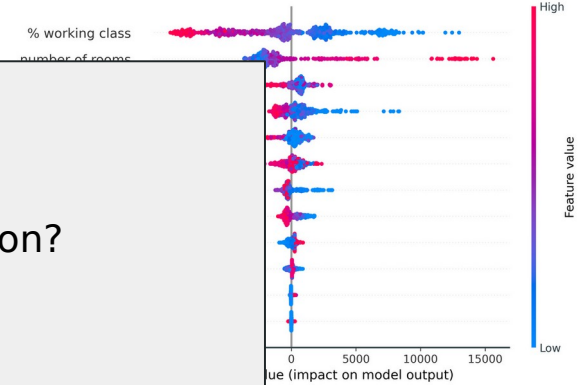
Counterfactual Illustration

Background

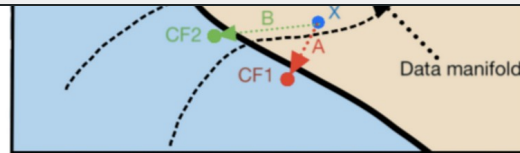


Lime Illustration

- 1 - Which one to choose for my application?
- 2 - Which one would a nurse\MD trust?
- 3 - How should a nurse\MD interpret the explanations?
- 4 - How would a nurse\MD use these methods?
- 5 - How would a nurse\MD interact with the method?
- 6 - ...

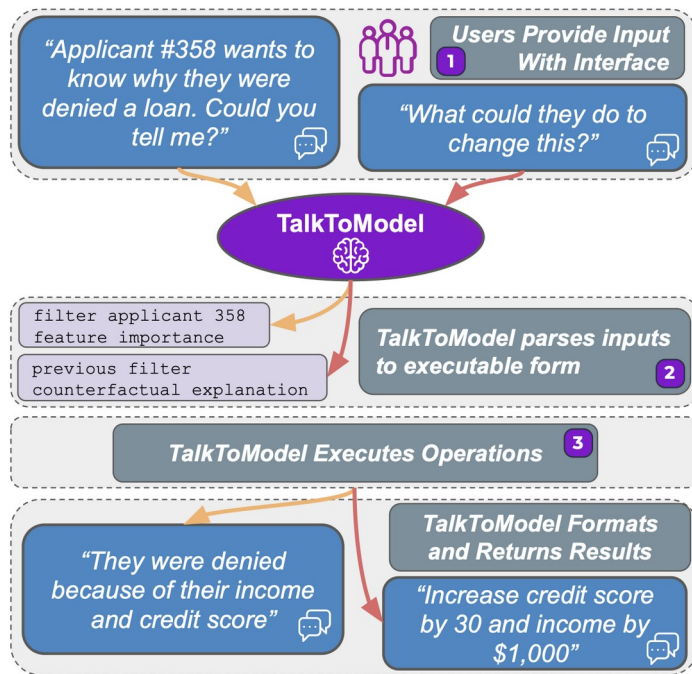


Illustration



Counterfactual Illustration

Talking to Model



1 - It uses **many** (with the possibility to add more) **post-hoc explanations**!

2 - **It chooses the "best"** explanation to practitioner!

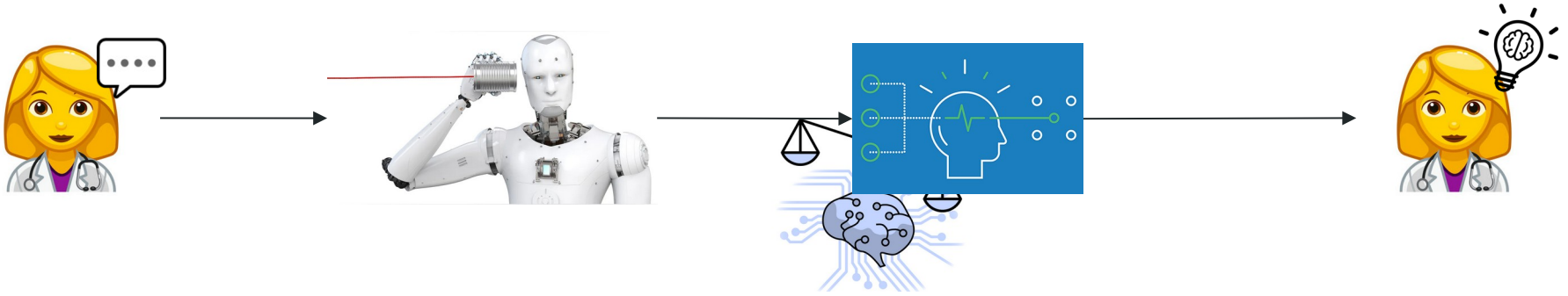
3 - It **answer** user's questions with **natural language**!

4 - Users **only** need to provide the **model and the data**!

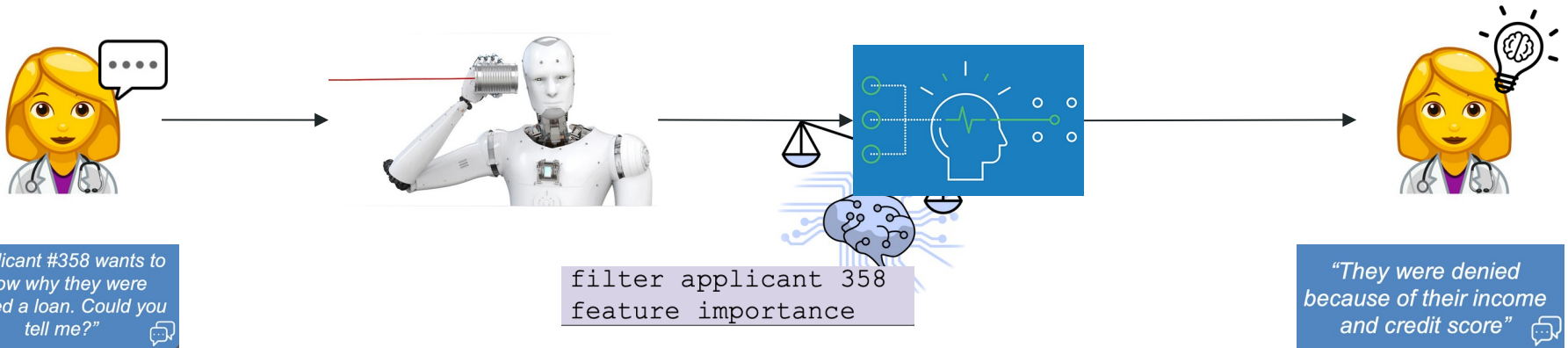
5 - They can **communicate via natural language**!

6 - ...

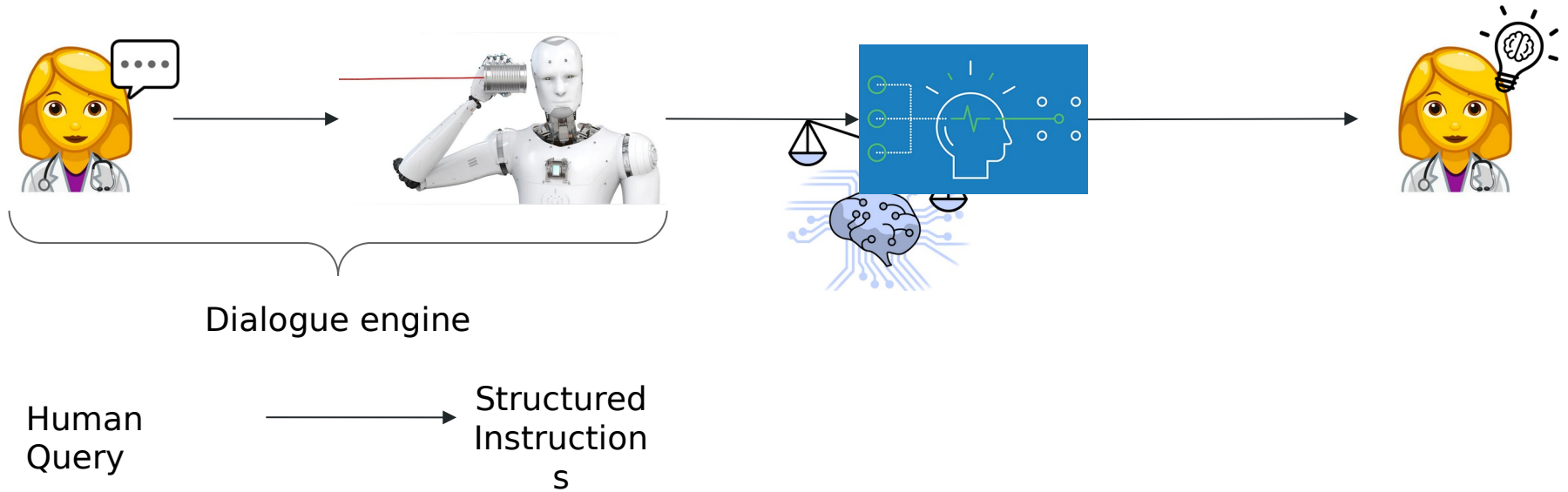
Method's Structure



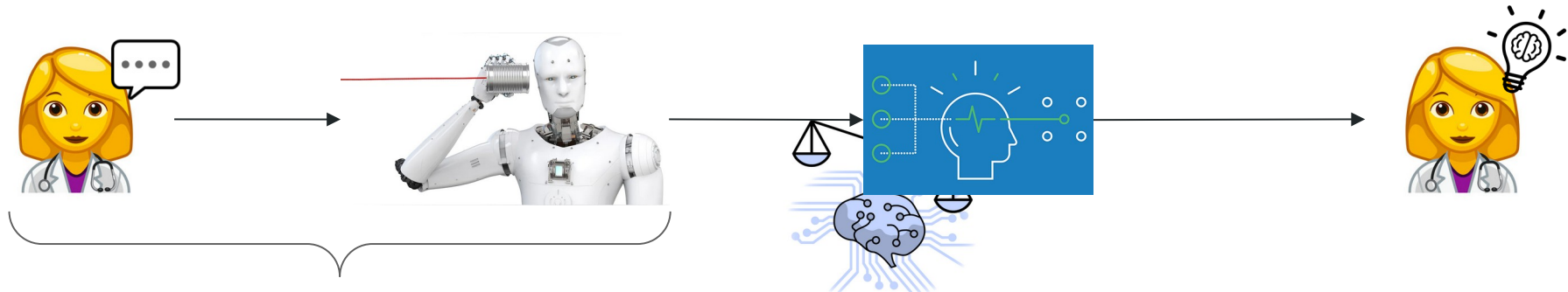
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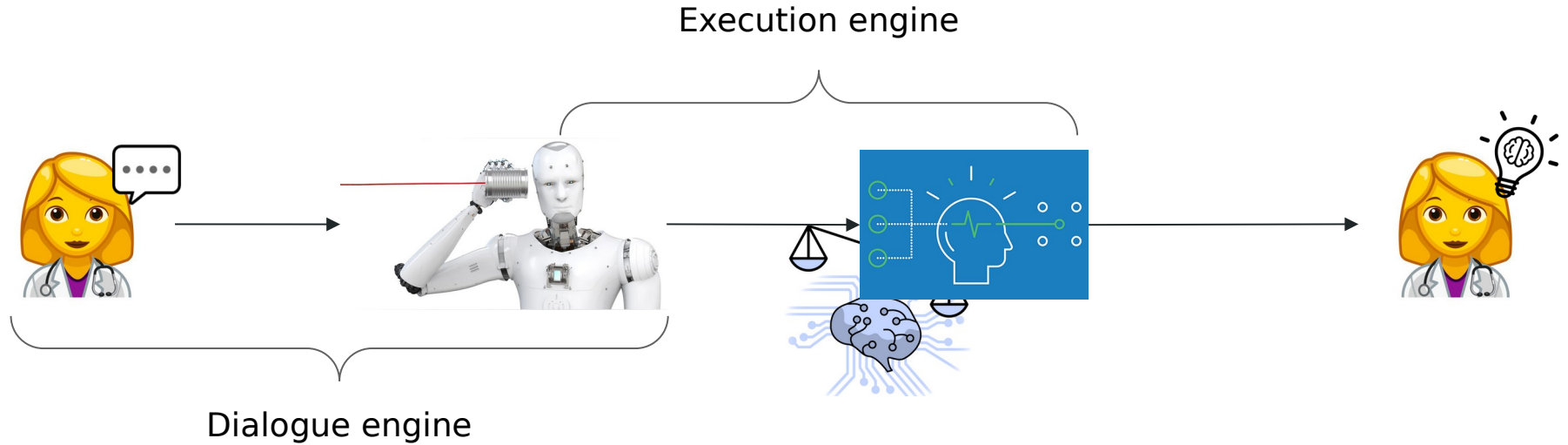


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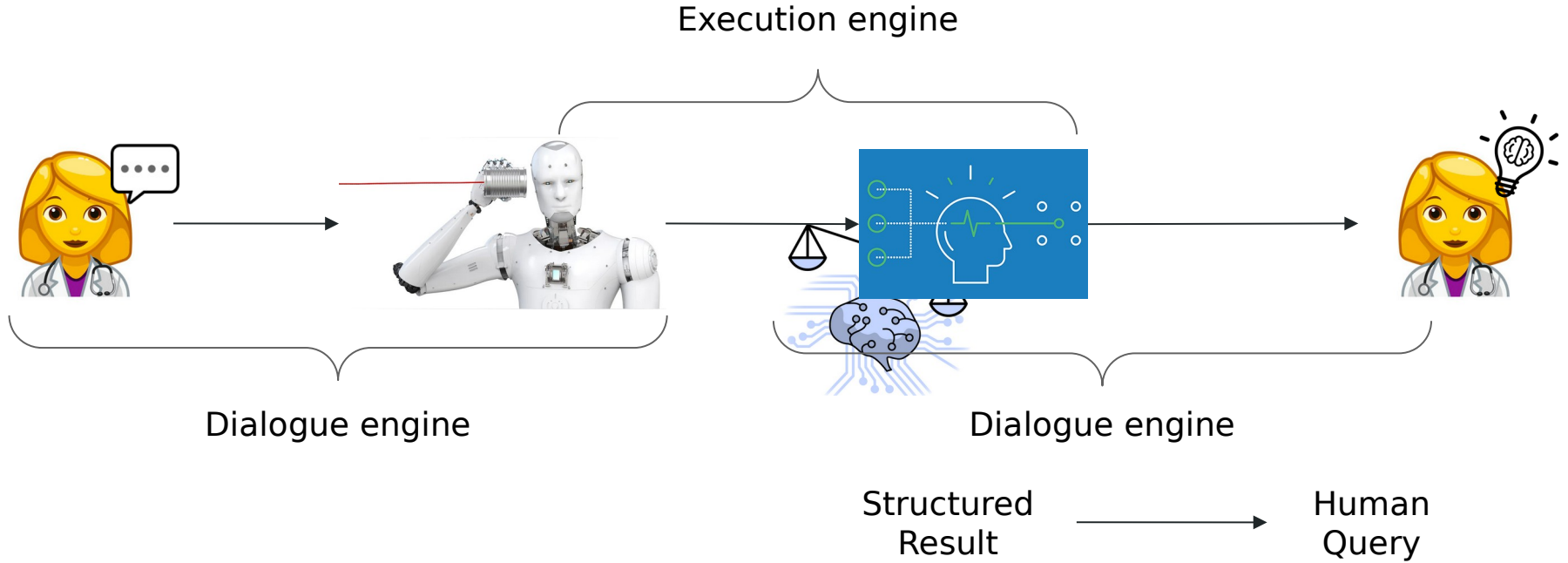


Human Query → **LLMs** → Structured Instructions

Method's Structure



Method's Structure



Dialogue Engine



Constructing a grammar

How is the grammar generated?

There is a predefined grammar that depends on the production rules that includes:

- 1 - All the operations that TalkToModel can run,
- 2 - The arguments for each operation,
- 3 - The relations between operations!

[illegible]

Constructing a gra



How to define a grammar for different dataset?

It is a challenge to use a general grammar that works for all dataset.



TalkToModel uses a grammar that is dependent on the dataset features!

3 - The relations between operations!

Dialogue Engine



Fine tuning LLM
Txt to Parses

“To parse user utterances into the grammar, we finetune an LLM to translate utterances into the grammar in a seq2seq fashion”

Dialogue Engine



Fine tuning LLM
Txt to Parses

What is finetune?

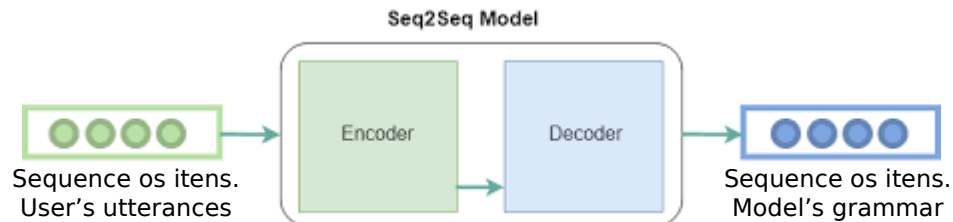
*“To parse user utterances into the grammar, we **finetune** an LLM to translate utterances into the grammar in a seq2seq fashion”*

Dialogue Engine



Fine tuning LLM
Txt to Parses

“To parse user utterances into the grammar, we finetune an LLM to translate utterances into the grammar in a seq2seq fashion”



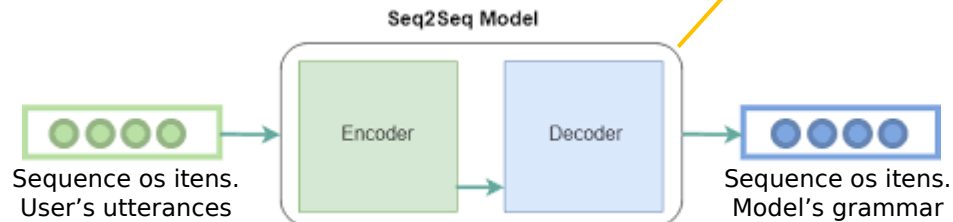
Dialogue Engine



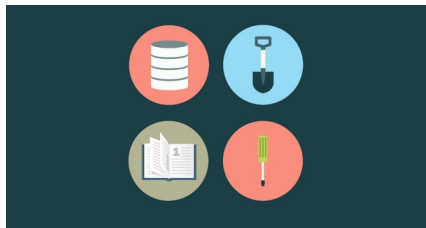
Fine tuning LLM
Txt to Parses

- Few-shot GPT-J
- Finetuned T5

“To parse user utterances into the grammar, we finetune an LLM to translate utterances into the grammar in a seq2seq fashion”



Dialogue Engine

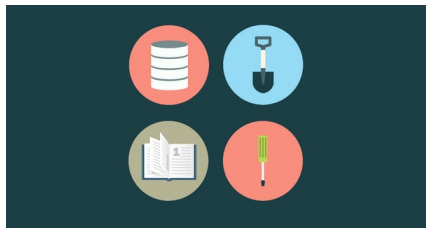


Generate fine tuning data

How to generate fine tuning data?

Human annotation?

Dialogue Engine

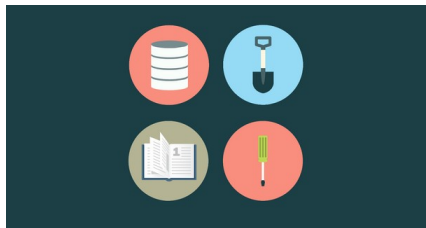


Generate fine tuning data

How to generate fine tuning data?

~~Human annotation?~~

Dialogue Engine

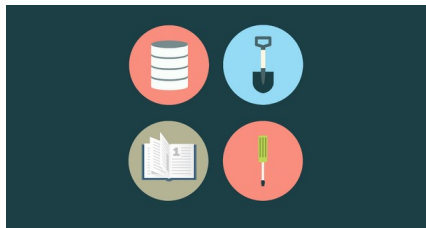


Generate fine tuning data

How to generate fine tuning data?

- 1 - Write a initial set of user's utterances and parses (where part of utterances & parses are wildcards terms).
- 2 - TalkToModel enumerates the wildcards with terms in the user's provided data

Dialogue Engine



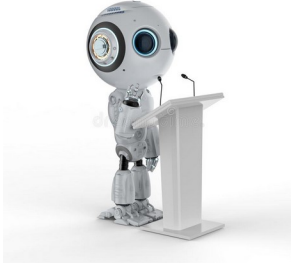
Generate fine tuning data

How to generate fine tuning data?

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TalkToModel usually generates 20k to 40k pairs of parses.

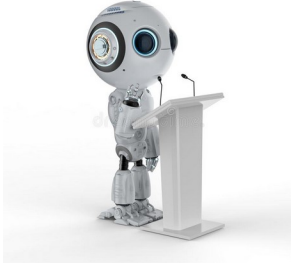
Dialogue Engine



Respond
conversationally

“After TalkToModel executes a parse, it composes the results of the operations into a natural language response it returns to the user.”

Dialogue Engine



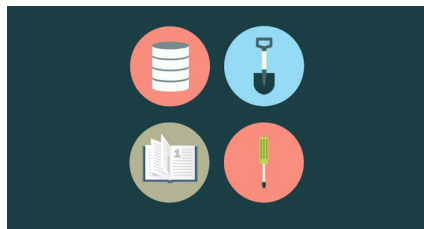
Respond
conversationally

- TalkToModel generates responses using **templates** associated with each operation!
- TalkToModel can run multiple operations at the same time. In this case, the model will join responses templates ensuring semantic coherence.

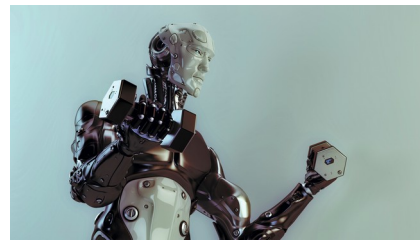
Dialogue Engine



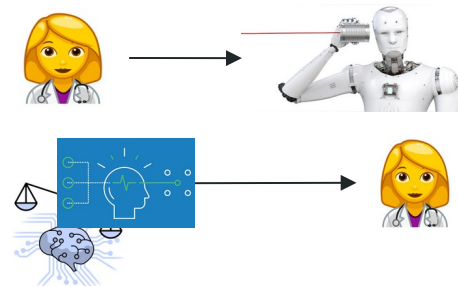
Constructing a grammar



Generate fine tuning data

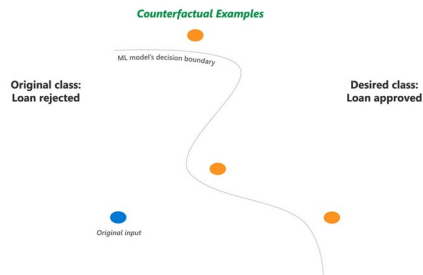
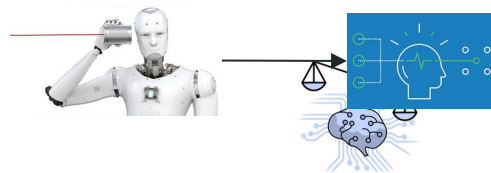


Fine tuning LLM
Txt to Parses



Respond
conversationally

Execution Engine



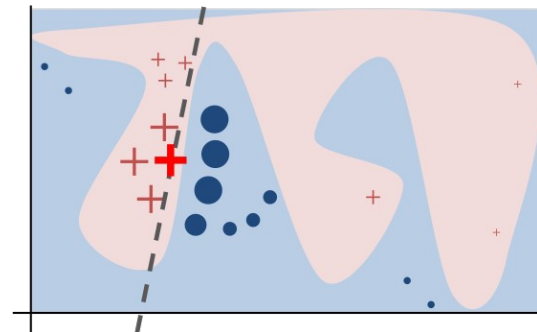
Counterfactual Explanations:

TalkToModel uses DiCE because it provides a diverse set of counterfactuals.



Data and Predictions Exploration

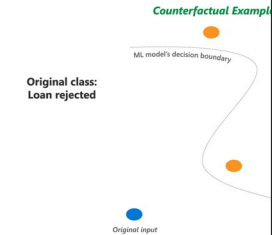
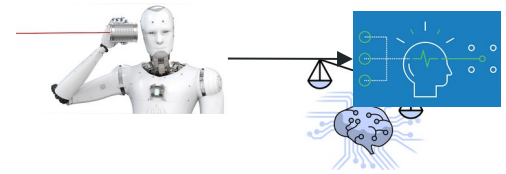
TalkToModel allow users to analyse predictions, inspect the model for errors, and analyse de data itself.



Post-hoc Feature Explanations

TalkToModel uses LIME and SHAP to make feature importance based explanations.

Execution Engine

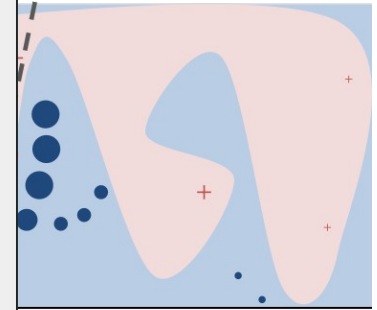


Counterfactual Explanation

TalkToModel uses DICE because it provides a diverse set of counterfactuals.

TalkToModel selects the best explanation!

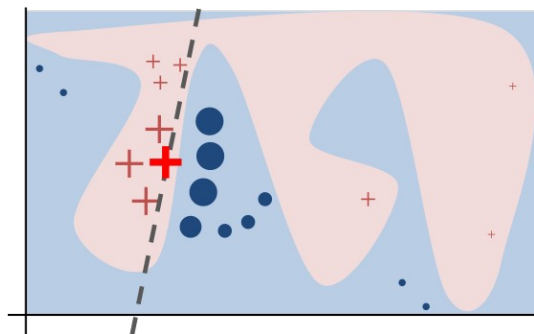
Instead of providing the LIME coefficients or the SHAP values, the model test the methods and provide the “best” feature based explanation.



Ad-hoc Feature Explanations

TalkToModel uses LIME and SHAP to make feature importance based explanations.

Execution Engine



Explanation Selection

How to select the best explanation?

Setup:

$$f(\mathbf{x}) \rightarrow \mathbf{y}$$

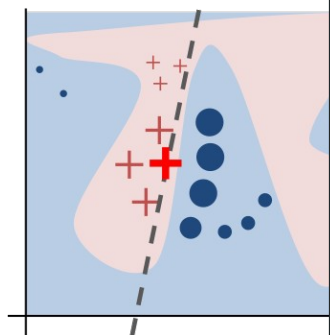
Is the model outputting the probability (y) of a class.

$$\Phi(\mathbf{x}, f) \rightarrow \phi$$

Feature importances for model f on the data $\mathbf{x}$. greater magnitudes correspond to higher importance features

Execution Engine

How to select the best explanation?



Explanation Set



How to choose the “best” feature importance?



More Important

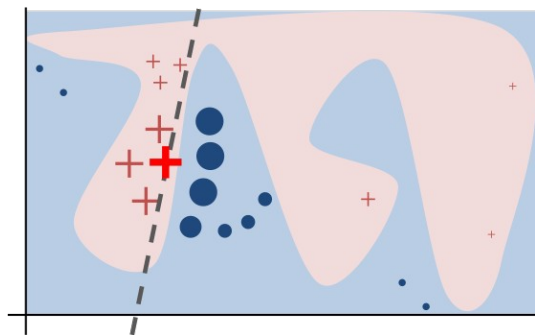
=

Small feature
perturbations
lead to
Big score perturbations

g the probability (y)

for model f on the
itudes correspond to
atures

Execution Engine



Explanation Selection

How to select the best explanation?

The Fudge score!

$$\text{Fudge}(f, \mathbf{x}, \mathbf{m}) = \frac{1}{N} \sum_{n=1}^N |f(\mathbf{x}) - f(\mathbf{x} + \epsilon_n \odot \mathbf{m})|$$

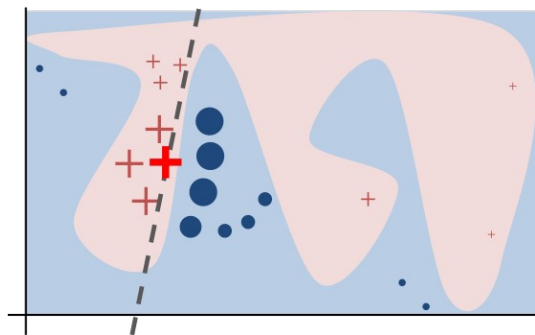
original
prediction

perturbed
prediction

Gaussian
Noise

Perturbed
features are
given by $\mathbf{m}$

Execution Engine



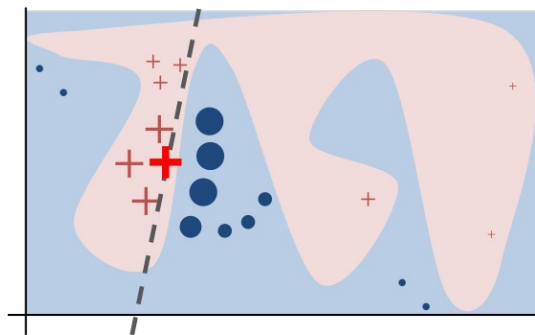
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Execution Engine



Explanation Selection

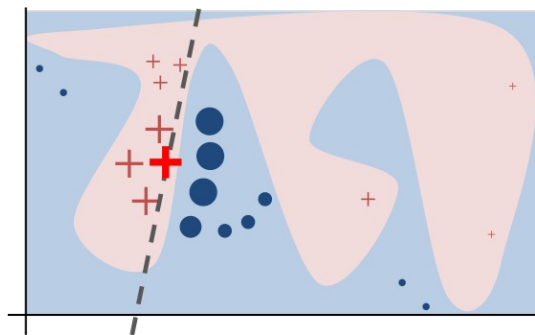
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Average Magnitude of
perturbations
by adding feature noise

Execution Engine



Explanation Selection

How to select the best explanation?

The Fudge score

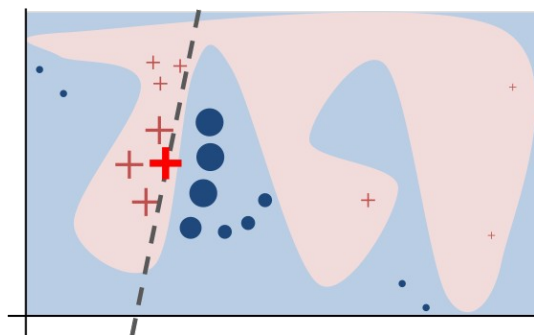
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Feature Importance Faith

$$\text{Faith}(\phi, f, \mathbf{x}, K) = \sum_{k=1}^K \text{Fudge}(f, \mathbf{x}, \mathbb{1}(k, \phi))$$

Indicator of top k features

Execution Engine



Explanation Selection

How to select the best explanation?

The Fudge score

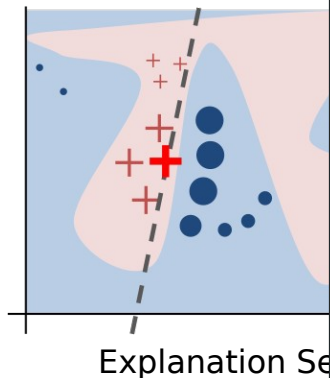
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Feature Importance Faith

$$\text{Faith}(\phi, f, \mathbf{x}, K) = \sum_{k=1}^K \underbrace{\text{Fudge}(f, \mathbf{x}, \mathbb{1}(k, \phi))}_{\text{Fudge Score of top } k \text{ features}}$$

Execution Engine

How to select the best explanation?



Choose the BEST

TalkToModel computes the faithfulness of:

- LIME with kernels [0.25, 0.5, 0.75, 1.0]

● KernelSHAP

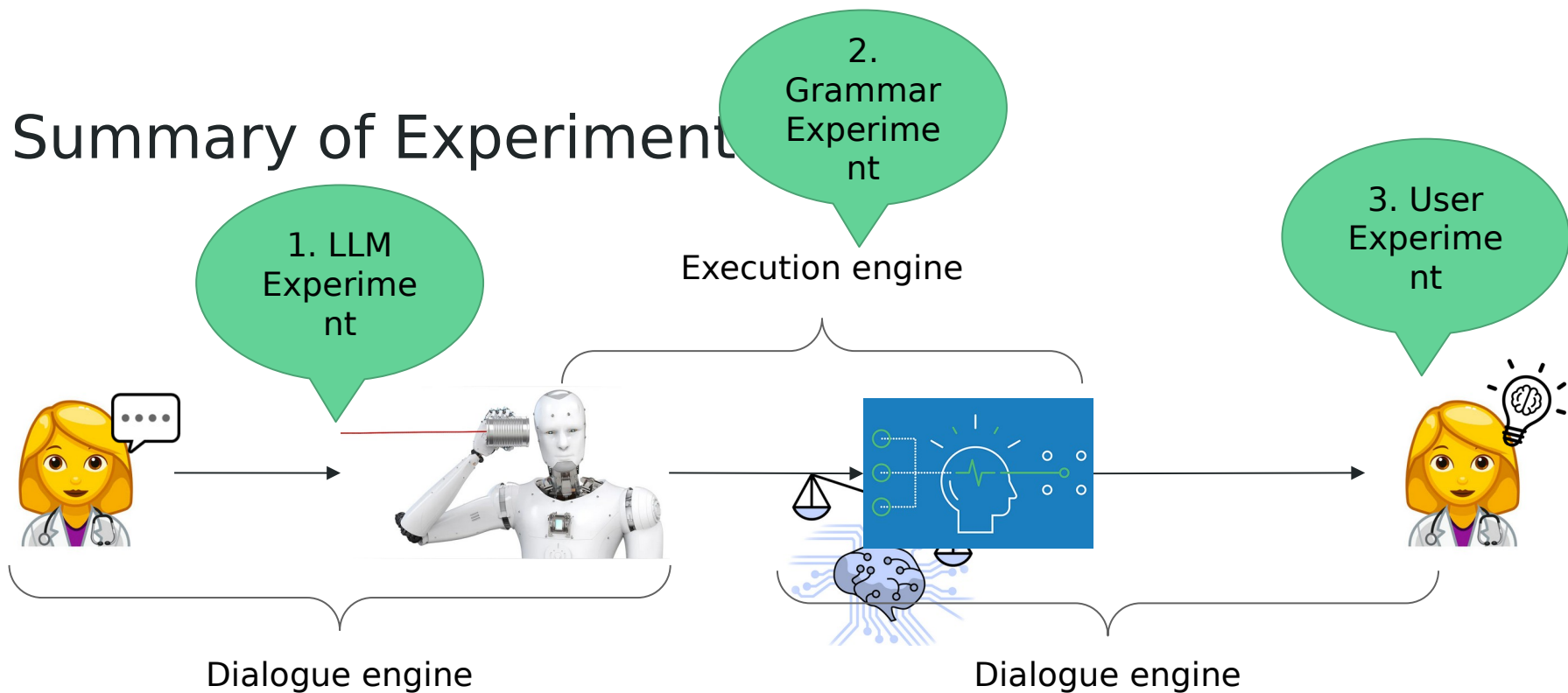
Report the one with highest faith!

$|\phi(\mathbf{m})|$

$|\phi(\mathbf{x}, \phi)|$

Results

Summary of Experiment



1. LLM Experiment

Is the LLM accurately interpreting the user's question?

- Create a “Gold Dataset” - ground truth (utterance, parse) pairs specific to one application domain
- Evaluate “Exact Match Accuracy” of LLM translation

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Compare along **easy** and **hard** splits of the data

Compare along **n-shot** and **fine-tuning** for different LLM sizes

Data Collection



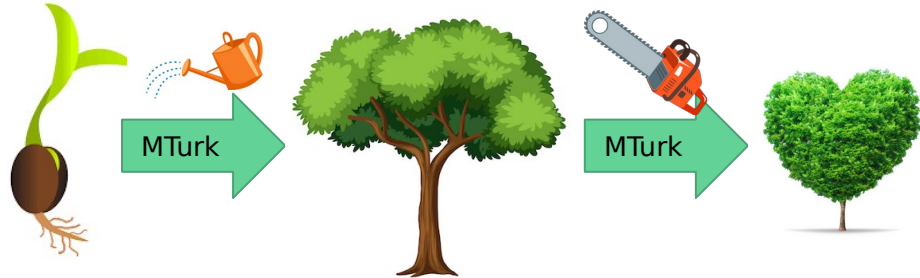
- Authors handwrote 50 (utterance, parse) pairs for each domain
 - Enforce that every operation appears at least twice for good coverage

Data Collection



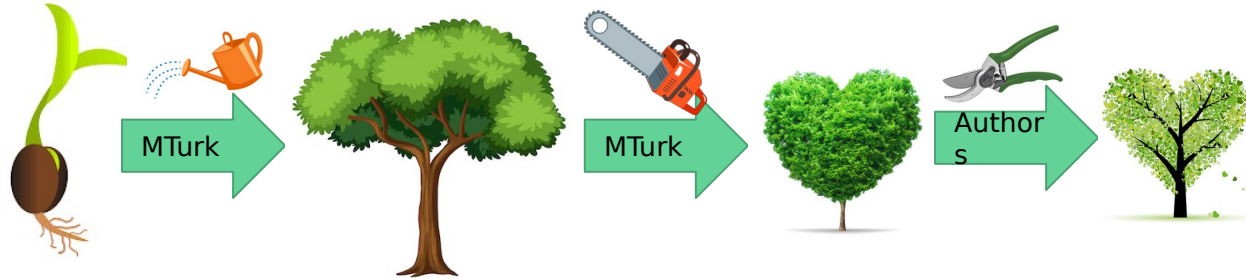
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 - For a total of 400

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 - Keep those that rate 3/4 or higher averaged over 5 raters
- Manual Filtering by Authors
 - Unclear (in paper) what proportion of filtering is done by MTurk vs. Authors

Example Paraphrases from TTM's Open Source Data



“What is your reasoning for determining if people older than 20 are likely to commit crimes?”

Example Paraphrases from TTM's Open Source Data



“What is your reasoning for determining if people older than 20 are likely to commit crimes?”



- “Why do you think people over the age of twenty are likely to commit a crime?”
4 4 4 3 3
- “How did you determine the likelihood of people over 20 committing crimes?”
2 4 3 2 4
- “Can you reason why people over twenty would likely commit crimes?”
3 2 4 3 2

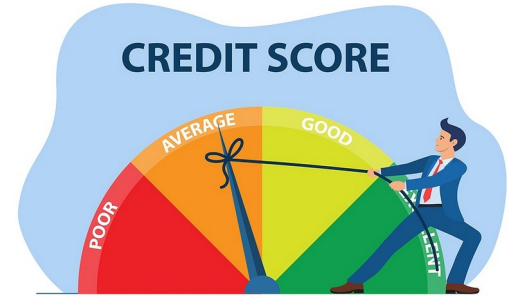


Domains/Datasets

- Diabetes: Pima Indian Diabetes Dataset



- Credit: German Credit Dataset



- Recidivism: COMPAS



Domains/Datasets

- Diabetes: Pima Indian Diabetes Dataset

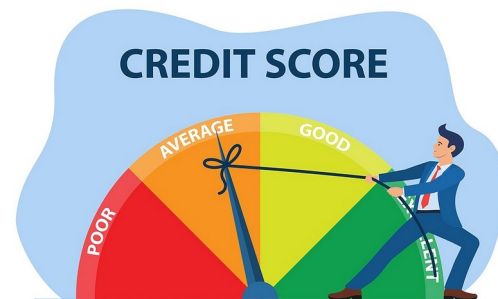
- 768 Women from Phoenix, AZ
- 8 Health Features, Diabetes or Not
- Questions: 400 → 190

- Credit: German Credit Dataset

- 1000 Loan Applicants
- 20 Financial Features, Good or Bad
- Questions: 400 → 200

- Recidivism: COMPAS

- 11757 Men and Women Criminal Defendants
- 43 Demographic/History Features, Risk Score 1
- Questions: 400 → 146



Splits of the Gold Dataset

IID (Easy)

- Order of operations are in the training data
 - Allowing different arguments



Compositional (Hard)

- Order of operations not seen before in the training dataset





	German			Compas			Diabetes		
	IID	Comp.	Overall	IID	Comp.	Overall	IID	Comp.	Overall
Nearest Neighbors	26.2	0.0	16.5	27.4	0.0	21.9	10.9	0.0	8.4
GPT-Neo 1.3B									
10-SHOT	41.3	4.1	27.5	35.9	0.0	28.8	40.1	7.0	32.6
20-SHOT	39.7	0.0	25.0	39.3	0.0	31.5	42.9	2.3	33.7
30-SHOT	42.9	0.0	27.0	39.3	0.0	31.5	41.5	4.7	33.2
GPT-Neo 2.7B									
5-SHOT	38.1	4.1	25.5	35.9	3.4	29.5	46.9	7.0	37.9
10-SHOT	38.1	6.8	26.5	40.2	3.4	32.9	40.8	9.3	33.7
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GPT-J 6B									
5-SHOT	51.6	14.9	38.0	51.3	6.9	42.5	55.8	7.0	44.7
10-SHOT	57.9	9.5	40.0	49.6	3.4	40.4	53.7	9.3	43.7
T5									
SMALL	61.1	32.4	50.5	71.8	10.3	59.6	77.6	30.2	66.8
BASE	68.3	48.6	61.0	65.0	10.3	54.1	84.4	34.9	73.2
LARGE	74.6	44.6	63.5	76.9	24.1	66.4	84.4	51.2	76.8

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2. Grammar Experiment

Is the grammar expressive enough to capture all XAI questions?

- Use an **XAI question bank**
 - Previous work, informed by design expert interviews
- Manually review if grammar can answer questions
- 30/31 questions can be answered!
 - More questions deemed out of scope

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What features does the system consider? `deemed on`
`topk(test_data, all)`

What would the system predict if a given feature A changes to..?
`predict(change(filter(test_data, id, A), feature, value, set))`

What kind of mistakes is the system likely to make?
`mistakes(test_data)`

How should instance A change to get a different prediction Q?
`cfe(filter(test_data, id, A, =), 10, Q)`

	operation, arguments, and description
Data	<code>filter(dataset, feature, value, comparison)</code>: filters <code>dataset</code> by using value and comparison operator <code>change(dataset, feature, value, variation)</code>: Changes <code>dataset</code> by increasing, decreasing, or setting feature by <code>value</code> <code>show(list)</code>: Shows items in list in the conversation <code>statistic(dataset, metric, feature)</code>: Computes summary statistic for <code>feature</code> <code>count(list)</code>: Length of list <code>and(op1, op2)</code>: Logical “and” of two operations <code>or(op1, op2)</code>: Logical “or” of two operations
Explainability	<code>explain(dataset, method, class=predicted)</code>: Feature importances on <code>dataset</code> <code>cfe(dataset, number, class=opposite)</code>: Gets <code>number</code> counterfactual explanations <code>topk(dataset, k)</code>: Top <code>k</code> most important features <code>important(dataset, feature)</code>: Importance ranking of <code>feature</code> <code>interaction(dataset)</code>: Interaction effects between features <code>mistakes(dataset)</code>: Patterns in the model’s errors on <code>dataset</code>
ML	<code>predict(dataset)</code>: Model predictions on <code>dataset</code> <code>likelihood(dataset)</code>: Prediction probabilities on <code>dataset</code> <code>incorrect(dataset)</code>: Incorrect predictions <code>score(dataset, metric)</code>: Scores the model with <code>metric</code>
Conv.	<code>prev_filter(conversation)</code>: Gets last filters <code>prev_operation(conversation)</code>: Gets last non-filtering operations <code>followup(conversation)</code>: Respond to system followups
Description	<code>function()</code>: Overview of the system’s capabilities <code>data(dataset)</code>: Summary of dataset <code>model()</code>: Description of <code>model</code> <code>define(term)</code>: Defines <code>term</code>

3. User Experiment

- Diabetes dataset with a gradient-boosted tree model
- 45 healthcare workers; 12 ML grads
- Answer 10 XAI MC questions
- Survey user preference vs. *explainerdashboard*
 - Ease of use, confidence, speed, and likability

Example question: “Is glucose more important than age for the model’s predictions for data point 49?”

Comparison	% Agree TalkToModel Better	
	Health Care Workers	ML Grad. Students
Easiness	82.2	84.6
Confidence	77.7	69.2
Speed	84.4	84.6
Likeliness To Use	73.3	53.8

	% Questions Completed		% Accuracy On Completed Questions	
	Dash.	TalkToModel	Dash.	TalkToModel
Health Care Workers	74.7	86.2	66.1	91.8
ML Grad. Students	73.8	93.9	62.5	100.0

explainerdashboard

Model Explainer

Positive class:

Survived x

Feature Importances	Classification Stats	Individual Predictions	What if...	Feature Dependence	Feature Interactions	Decision Trees
---------------------	----------------------	------------------------	------------	--------------------	----------------------	----------------

Feature Importances

Which features had the biggest impact?

Importances type:

- ☐ Permutation Importances
☒ SHAP values

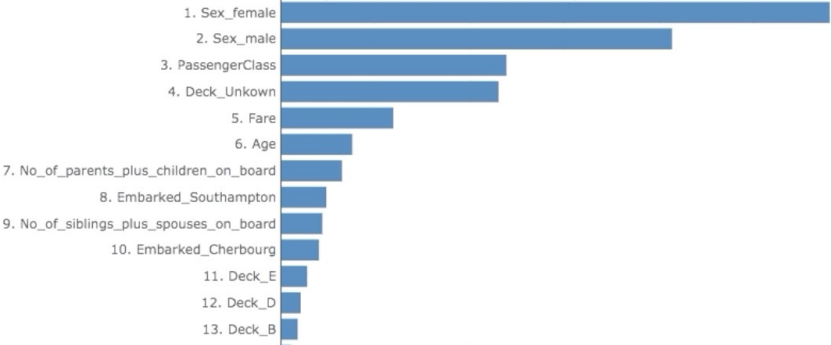
Depth:

Select...

Grouping:

☒ Group cats

Average impact on predicted Survival
(mean absolute SHAP value)



TalkToModel

how likely are patients older than forty to have diabetes?

Over 85 cases where **age greater than 40.0** in the data, the model predicts:

- **unlikely to have diabetes**, 44.706% of the time
- **likely to have diabetes**, 55.294% of the time



why?

For instances with **age greater than 40.0** predicted *likely to have diabetes*:

- **glucose** is the **most important** feature and has a *positive* influence on the predictions
- **bmi** is the **second** most important feature and has a *positive* influence on the predictions
- **age** is the **third** most important feature and has a *positive* influence on the predictions

For instances with **age greater than 40.0** predicted *unlikely to have diabetes*:

Enter your command! Use the ↑ arrow and ↓ arrow to cycle previous command

Send

 Help me generate a question about... 

Pins

Pin messages and they will appear here!

There are **85 items** in the data where **age greater than 40.0**.
Let me know if you want to see their ids.

For **all** the instances in the data, the statistics of **age** in the dataset are:
mean: 34.192
one std: 12.146
min: 21
max: 81

Findings

Users using TalkToModel (as compared to explainerdashboard)

- Over 90% accurate compared to 60% otherwise
- Got answers in half the time
- Consistently preferred it on ease of use, confidence, speed, and likability



Conclusions



- TalkToModel provides an **elegant UI** and conversation tool that makes interpreting models **easier** for laypeople and ML practitioners alike
- **Highly extensible** to handle a variety of explainability needs, problem domains, and XAI methods
- **Reasonably accurate** at interpreting user intent and provides well-formatted templated responses
- A dataset of your own is all you need to use TalkToModel



Limitations

- Authors do not test the system in **real world settings**
- Automation of explainability methods gives **no flexibility** on user side (e.g., “what is the most feasible CFE” or “what is the most stable post-hoc explanation”)
- No guarantees on **data quality** for both training and fine-tuning, and still requires some manual labor
- System possesses no **domain knowledge** besides grammar
- **Accuracy on hard split** is still very low

Discussion Questions

- Are ML practitioners the most responsible for accessibility of XAI?
- Is the TalkToModel LLM itself interpretable?
 - When is it acceptable to improve XAI with more black box AI?
- How much control over the explanation should we give to the user while still being accessible to the layperson?
- Does the existence of TalkToModel excuse the need for other XAI methods to be accessible?
- Dashboard vs. Dialogue?

